BSc (Hons) Software Engineering - Coventry University
National Institute of Business Management (NIBM)

NB6007CEM - Web API Development

Seed Data and Deployment

Day 1 (PM) - Session 2

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Where We Left Off

Domain model agreed:

- Province - District - PoliceStation - Vehicle - LocationPing + User
- device_id is a Vehicle attribute - no separate Device entity
- LocationPing is a time-series - not a field on Vehicle
- Design pipeline: Step 1 (data model) complete - eight steps ahead
- Boundary: no URLs, no JSON/XML choice, no endpoints today

Data generated today is not served yet - it waits in seed.json until endpoints are built on Day 2.

Today's Three Outputs

1
seed.json
All five entities

1
Express app
Single GET /

1
Public URL
Returns 200 on Render

Task	Time	Done when...
Task 1 - Seed Data	0:15 - 1:00	seed.json committed to repo
Task 2 - Hello-World App	1:00 - 1:45	GET / returns expected JSON locally
Task 3 - Deploy to Render	1:45 - 2:45	Public URL returns 200 OK
Checkpoint + Micro-Viva	**2:45 - 3:00**	**All three outputs verified**

Task 1 - Generate Seed Data

Time: 0:15 - 1:00 (45 min)

- **Open your AI assistant**
- **Paste the full prompt from the panel on the right**
- **Save output as seed.json in your project root**
- **Verify FK consistency (see next slide before committing)**
- **git add seed.json**
- **git commit -m "S2: seed data"**

SEED DATA PROMPT - PASTE IN FULL

Generate seed data for a Sri Lanka Police Tuk-Tuk Monitoring System as a single JSON object with five keys: provinces, districts, stations, vehicles, and pings.

Requirements:
- 9 provinces: id, name
- 25 districts: id, name, province_id
- At least 20 stations: id, name, district_id
- At least 200 vehicles: id, registration_number, device_id, station_id
- At least 7 days of pings per vehicle: id, vehicle_id, latitude, longitude, timestamp

CRITICAL: Every foreign key must reference an id that actually exists in the parent list. No orphaned records.

Seed Data - What Must Be True

FK chain - every foreign key must reference an existing parent id:

- **district.province_id must exist in provinces**
- **station.district_id must exist in districts**
- **vehicle.station_id must exist in stations**
- **ping.vehicle_id must exist in vehicles**

- **Scale: 9 provinces, 25 districts, 20+ stations, 200+ vehicles, 7+ days of pings per vehicle**

- **Timestamps are sequential and varied - not all the same value**

What generators get wrong: Flat unrelated lists with broken FK chains - e.g. district_id: 99 referencing a district that does not exist. Always verify before committing.

Task 2 - Express App

Time: 1:00 - 1:45 (45 min)

- Paste the fallback prompt (right) into your AI assistant
- Save output as index.js and package.json
- npm install
- npm start
- Test: GET http://localhost:3000/
- Verify response: {"status":"ok","session":"NB6007CEM S2"}
- git add . && git commit -m "S2: hello-world app"

FALLBACK PROMPT - USE THIS VERBATIM

"Generate a minimal Express.js app with a single GET / route returning JSON {status:'ok', session:'NB6007CEM S2'}. Include package.json with a start script using node. No database, no other routes, no middleware."

Before deploying to Render, ensure your app uses:
app.listen(process.env.PORT || 3000)

If generation produces broken or incomplete code, use the fallback prompt above unchanged.

Task 3 - Deploy

Time: 1:45 - 2:45 (60 min)

- **Push your repo to GitHub (if not already done)**
- **Go to render.com - sign up with GitHub account**
- **New > Web Service > Connect your repository**
- **Build command: npm install**
- **Start command: node index.js**
- **Instance type: Free**
- **Click Deploy - watch the logs for "Listening"**
- **Copy public URL and share with instructor**

CRITICAL - PORT BINDING

Render injects the port via an environment variable.

Your listener MUST be:
app.listen(process.env.PORT || 3000)

Without this the build succeeds but the service fails to start.

FREE TIER NOTE

Free services sleep after 15 min of inactivity.
First request after sleep takes up to 30 sec.
This is normal for development work.

Common Failures - What to Check

Failure	What to Check
Port binding fails on Render	Must use process.env.PORT. Render assigns the port dynamically. A hardcoded port 3000 will cause the service to fail on startup.
Start command mismatch	The start command in Render must match the actual entry file name. index.js, app.js, and server.js are different files.
seed.json FK errors	Manually verify: does every district_id in stations exist in the districts list? Regenerate with the same prompt if the chain is broken.
git push fails or not linked	The repository must be pushed to GitHub before linking it in Render. A local git commit alone is not visible to Render.

Day 1 Checkpoint

1. Public URL returns 200 with {"status":"ok","session":"NB6007CEM S2"}
2. seed.json committed to repo - all five entities, FK chain intact
3. Repository shared with instructor as a GitHub collaborator

All three must be true. Localhost is accepted only if Render is confirmed down - document it in your README and resolve before Session 3.

Day 1 Micro-Viva

1. Point to one entity in your model and explain why it is there and what it owns.
2. Why is device_id an attribute of Vehicle rather than a separate Device entity?
3. What is the difference between the data model and the resource model?
4. What URL is your API deployed at? Show me it returns 200.
5. Why does the seed data need FK consistency before we write a single endpoint?

Questions & Discussion

Day 1 done - commit, push, and share your repo before you leave.

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